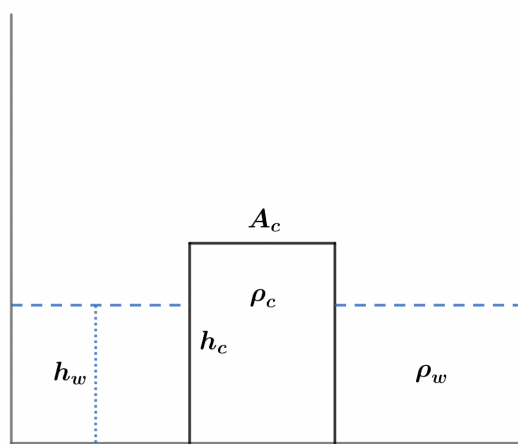


2022A F=ma Exam: Problem 18

Kevin S. Huang



When the cork starts to float, its weight is balanced by the buoyant force,

$$\begin{aligned}F_g &= m_c g = \rho_c A_c h_c g \\F_B &= \rho_w V_{sub} g = \rho_w A_c h_w g \\ \rho_c A_c h_c g &= \rho_w A_c h_w g\end{aligned}$$

Solving for the water height,

$$h_w = \frac{\rho_c h_c}{\rho_w}$$

so the answer is B.